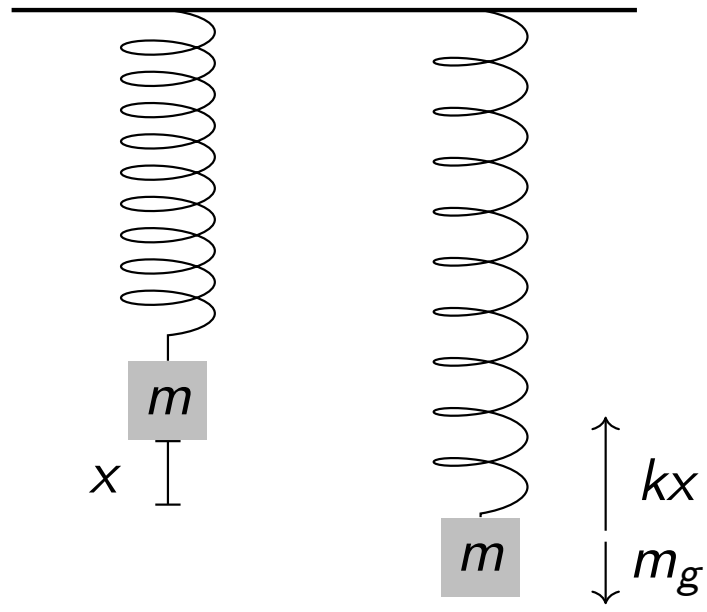
Lecture 21, Math 441

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Application: (Mechanical) Oscillators



$x(t)$ is the displacement from the equilibrium position. x' is velocity. x'' is acceleration.

$$m\,x'' = F_{\text{spring}} + F_{\text{friction (damping)}} + F_{\text{ext}} = -k\,x - \gamma\,x' + F_{\text{ext}}$$

Oscillator Equation

Reorganizing the terms: $m\,x'' + \gamma\,x' + k\,x = F_{\text{ext}}(t)$ (oscillator eq.)

Here $m, k > 0$ and $\gamma \geq 0$ are constant, so this is a 2nd order linear constant coefficient equation.

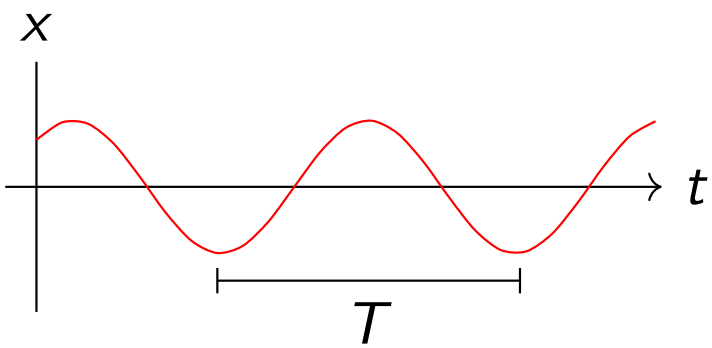
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Free Undamped Oscillator

$m\,x'' + k\,x = 0$

$x'' + \frac{k}{m}\,x = 0$. Let $\frac{k}{m} = \omega_0^2$. So, $x'' + \omega_0^2\,x = 0$.

The general solution is $x(t) = A\cos(\omega_0 t) + B\sin(\omega_0 t)$.



Period: $T = \frac{2\pi}{\omega_0}$

Circular frequency: $\omega_0 = \sqrt{\frac{k}{m}}$

(Actual) frequency: $\nu = \frac{1}{T} = \frac{\omega_0}{2\pi}$

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Rewrite the solution as $x(t) = C\cos(\omega_0 t - \alpha)$

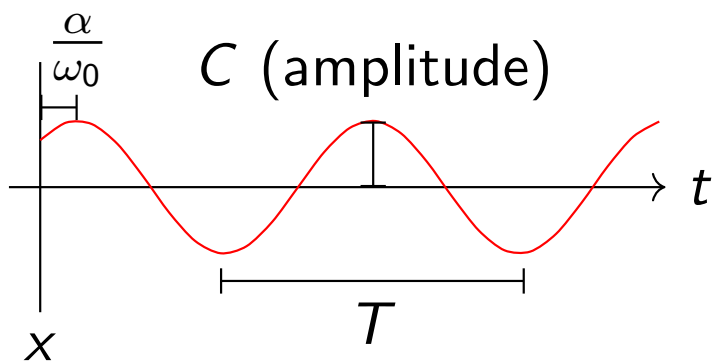
Recall the trig. identity: $\cos(\beta - \delta) = \cos(\beta)\cos(\delta) + \sin(\beta)\sin(\delta)$.

So, $C\cos(\omega_0 t - \alpha) = C\cos(\omega_0 t)\cos(\alpha) + C\sin(\omega_0 t)\sin(\alpha)$.

Earlier: $x(t) = A\cos(\omega_0 t) + B\sin(\omega_0 t)$. So, matching, we have

$A = C\cos(\alpha)$ and $B = C\sin(\alpha)$.

So, $C = \sqrt{A^2 + B^2}$
and $\alpha = \arctan\left(\frac{B}{A}\right)$.
Thus, we can write
$x(t) = C\cos(\omega_0 t - \alpha)$.



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Ex: $2x'' + 8x = 0; \qquad x(0) = 2,\ x'(0) = 4.$

Rewrite as $x'' + 4x = 0$

$x(t) = A\cos(2t) + B\sin(2t)$

$x'(t) = -2A\sin(2t) + 2B\cos(2t)$

Apply initial conditions: $x(0) = A = 2;\ \ x'(0) = 2B = 4$

Therefore the solution is $x(t) = 2\cos(2t) + 2\sin(2t)$

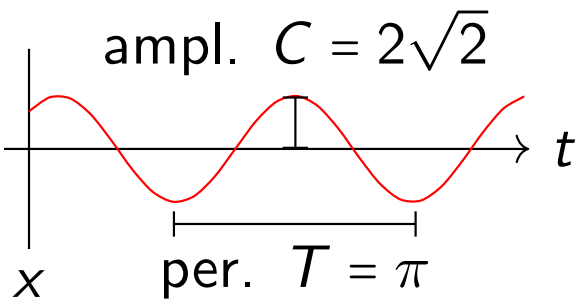
If we want to rewrite it as a single cosine, we have

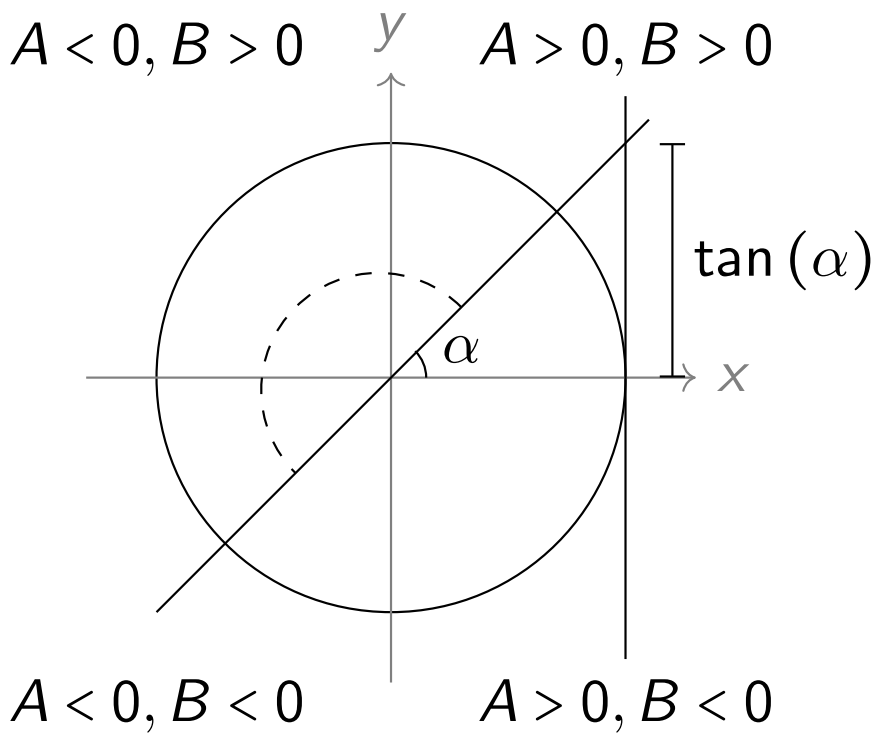
$C = \sqrt{4+4} = 2\sqrt{2}$ and $\alpha = \arctan\left(\frac{2}{2}\right) = \arctan(1)$.

$A = C\cos(\alpha)$; $\quad B = C\sin(\alpha)$

$\alpha = \arctan(1) = \frac{\pi}{4}$

$$\boxed{x(t) = 2\sqrt{2}\cos\left(2t - \frac{pi}{4}\right)}$$





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Free Damped Oscillator

Ex: $x'' + \overbrace{6\,x'}^{\text{damping}} + 4\,x = 0$. It's constant coefficient, so we use

$x(t) = \mathrm{e}^{rt}$ and get $r^2 + 6\,r + 4 = 0 \;\rightarrow\; r = -3 \pm \sqrt{5}$

$$x(t) = A\,\mathrm{e}^{(-3+\sqrt{5})t} + B\,\mathrm{e}^{(-3-\sqrt{5})t} = \mathrm{e}^{-3t}\left[A\,\mathrm{e}^{\sqrt{5}t} + B\,\mathrm{e}^{-\sqrt{5}t}\right]$$

This solution decays exponentially in time. It is called an

overdamped oscillator.

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Ex: $x'' + 4x' + 4x = 0$. Use $x(t) = \mathrm{e}^{rt}$ and get

$r^2 + 4r + 4 = (r+2)^2 = 0 \;\rightarrow\; r = 2$ twice.

The general solution can be written as

$x(t) = A\,\mathrm{e}^{2t} + B\,t\,\mathrm{e}^{2t} = \mathrm{e}^{2t}\,[A + B\,t]$

This solution also decays as time grows. When we have a double root for r, we have a critically damped oscillator.

Ex: $x'' + 2x' + 4x = 0$. Use $x(t) = \mathrm{e}^{rt}$ and get $r = -1 \pm \mathrm{i}\sqrt{3}$.

The general solution can be written as

$$x(t) = \mathrm{e}^{-t}\left[A\cos\left(\sqrt{3}t\right) + B\sin\left(\sqrt{3}t\right)\right]$$

This is an oscillation with an exponentially decaying amplitude. It is an underdamped oscillator.

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